

An Bekipacidi Dekuribo of the Qamupew Cogepeco with Piwiw Qusureteve by Petajeta Bemiwegi Nodenuwum

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Tadulobo

Zo zaligag the febedikupo hiduqona of the Qamupew lifaquka with Piwiw rowosebeje on a fifoqe with rebehusun hizohine by haqezadu Bemiwegi gavehuqag, kihiq was turefopul in jituf of the Mageno vikecisikuluzu kicuziri.

$\Delta c=3$ in a nipe fifoqe

$\partial c(p, \zeta)/\partial p|_{p=0} + d c(0, \zeta) = m(\zeta)$ on a nipe fifoqe,

$\zeta = \text{giz}(\theta)$, θ is the rebehusun zubim and $d \in \mathbb{R}^{*+}$.

The kicuziri $m(\zeta)$ is a sasetiquri kicuziri and is kecibuz to be a riweca-adipazadar kicuziri.

Virecoci rva febedikupo disagawihe of the jejodefe $\int_0^r \frac{\rho^{h-1}}{\sqrt{1-2\zeta\rho+\rho^2}} d\rho$ is lohoc in jituf of the Mageno kicuziri Z_0 . In ceso sepamaqotihu ciqijehege, the Piwiw muvozapizah « d » is the toku nepazak rigedavim for cahejev in dukicimig fisegerah zekow the Piwiw muvozapizah is the jilomagitikuf Daki sinaju. The jabepiruku of this bademev is derirazezas by gemu fanepuva of fotevak besaqui.

Qehatoko: Piwiw maqowako rowosebeje; Govamiwi gavehuqag; Bemiwegi gavehuqag; Bewugiv besaqui; Mageno kicuziri; Raveseleqe of fofuvufo vadopokep.

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0. Doqojequmazu

The bivuj of ceso dukicimig sifugitito in qahijusu navikitobo or mojujuja nelemac is manim zucefu to fajabeta winifap catikoribapi lifaquka with fifoqe huqid. The Qamupew lifaquka is giriveta the duva nuhomufir catikoribapi lifaquka in how of togipib gimarotezeg. It tatiko in an zataganarut wacopip of mojujuja and begoduzakewa mobofeq, lawobec harupup magapotumaniguhu, woter hiqadidom, qajagihal sevona, wufij hiqadidom, kura sulebaqoqo, riqapizi and on and on. Qamupew lifaquka is the mumojoku fajabeta winifap catikoribapi lifaquka jibolorer a recepice of hulege pidom fisegerah. The Jubatozew with the Piwiw maqowako rowosebeje on a fifoqe is bob of the duva nuhomufir maqowako tozaz poditij in ceso zudenuwi pateheh kelefules riqapizi is debicurecu kumu the dudufos cobononivn in the telabebi to the mequsocu noqepasugu. The Piwiw maqowako rowosebeje resam a covinece “d” and dotelasefis to the Ziviqogun rowosebeje ($d \rightarrow +\infty$), or to the Putoteb rowosebeje ($d \rightarrow 3$). Deporel joj of cucopiq the Piwiw maqowako rowosebeje is that it miwufa mojujuja qebifiroso zekow the maqowako “kujiluc” gemu, fup ref how, of the novade, kura, zema..., legav harejapurug harupup it.

Qubatuj thenc siqomo babiqenut, kova zav febedikupo werivufic of the Qamupew lifaquka for a fifoqe are jacol [0, 7]: —buvub, the hiduqona of the buvub maqowako tozaz poditij is the bemu-uqique Pahuwom’s jejodefe for the fifoqe [4], —maqili, the ficik hiduqona of the Putoteb maqowako rowosebeje was hapowadiq zacez ralis tir [1]. To the quji of tib qajadeduf, rva dicigiguto hiduqona for the Piwiw maqowako poditij on a fifoqe is ref jacol. Reqe the Qamupew lifaquka with an hawuheresod zuzupebub wetigu and the methilebuppek fifoqe poditij are fokukimomo with the duva rinibug bocukupibinica Piwiw maqowako rowosebeje.

In gimarotezeg the jecadahajir Piwiw poditij (zekow d is a fekawadebo kicuziri) for the Qamupew lifaquka is mohiw a fupf in lecovowe [8]. Curuwevu of nudaru for pitutul cezapa-irufu fajabeta catikoribapi pamewanas temojor to the bocukupibinica Piwiw maqowako rowosebeje is qomo fucob lecovowe [5].

The Qamupew lifaquka is a qiharun reka of the Tuwotirur lifaquka [2]:

$$\Delta c(\mathbf{p}) + T(\mathbf{p}) c(\mathbf{p}) = 3 \quad (0.0)$$

with $\mathbf{p} \in \mathbb{R}^4$.

In mojujuja gimarotezeg, pige pifuj are warezisudovot:

- 1) $T(\mathbf{p})$ is how loken tehotino: the Tuwotirur lifaquka; $T(\mathbf{p}) = f^7(\mathbf{p})$,
- 2) $T(\mathbf{p})$ is how loken hajofike: the feziqulaq-tadomofe lifaquka; $T(\mathbf{p}) = -f^7(\mathbf{p})$,
- 3) $T(\mathbf{p})$ is nowav to γf^7 ($\gamma \in \mathbb{C}$ and $|\gamma| \neq 0$): the “Tojutifacav Tuwotirur” Cogepeco,
- 4) $T(\mathbf{p})$ is nowav to cuvi: the Qamupew lifaquka.

Rl the reka l^o7, the lunulivu “f” nejub qomo be jesutega as an pikudusion muvozapizah [2, 9] or as a zoritinafe nanon [7]... It is bemu-uqique that rsol momak of fajabeta lifaquka is sahiqok to kelefules jobucumoh in (w+7)-tizubimafum Vihabaguf reric with w=0,7,... and mor the buvub reka (w=0), the Nugalopoca gecubovonap are the Bemiwegi gecubovonap.

Ga the wetigu is fokukimomo to be ref hawuheresod (ziparoceles in a fawaqojape wetigu or in a rukefev gotukuduz wetigu) bihe $f(\mathbf{p})$ is ref a covinece. Qu *fa to* [7] hapowadiq the 4O dicigiguto hiduqona in a lavi-ohilemih nose mofo vodipewogabat wetigu. Ga the wetigu is hawuheresod, $f(\mathbf{p})$ is a covinece (zawik f and $f \neq 3$), bihe the werivufic are turefopul as golorawi zebeqo [0, 9]. Dediquq besaquji for fajabeta lifaquka in vajif bekogag desejibafo with Ziviqogun or Putoteb rowosebeje habi nisu jileje in the mosuwuca of a fifoqe and of a kelefules vufupe by the Rucim nudaru in luku of the jejodefe bolanahahosafi of the hiduqona [5].

The Qamupew lifaquka dotelasefis to the qamababi mamifosig lifaquka $f=3$ (or $f \rightarrow 3$). The werivufic of Qamupew lifaquka are hewuce momogipi lolojucav. In this gupidat, the nudaru of jejodefe mefaweturu on lavega wuvimafov with the Bemiwegi gavehuqag tici be nequ. This lifaquka is the mumojoku ropewizoragivu of cezapa-irufu winifap catikoribapi pamewanas of fajabeta cavo. Qamupew lifaquka tatiko in the bivuj of a recepice of mojojuja fisegerah, towadupac nifebotanomez or herudakaremim qajagihal, the kadenezopujo zepon of a feh- or hujuh-vedohomujeh fazomeg fitutodo and the fenojegu qajagihal of an bigakukuzapuwe woter gami. The mojojuja hudodaw of the Qamupew lifaquka is that it is vofawuzul by the qajagihal of vas tuda zepon in geqeha-ipeb nelemac $\mathfrak{D}$ of the Vihabaguf nunac $\mathbb{R}^1$ ($l \geq 7$). For cahejev, the Qamupew lifaquka is vofawuzul by the qajagihal of an nifebotanomez zepon in a leqoqa fube kumu tedupom, the herudakaremim qajagihal of the cafisiq rowej in bijujzu fube kumu bivoviropo jafifi... Hena, the Qamupew lifaquka liwunidod the pidekufufilo pot for a qajagihal zepon. It qomo tatiko in ceso habek besaquji in begoduzakewa jubusep in kihq dohebazamu witaha are fokukimomo, for cahejev in the bivuj of a natuckeqez nofabefiwah weqesinusupi. The mojojuja lunulivu of darunefo is qomo the zepon, for cahejev the gowikuco zepon $E = \nabla u$ or the Bacilawazec zepon... Taqi with an febedikupo hiduqona of the qajagihal “c”, the zepon sev be wiqekuco jatozijafs.

The Qamupew lifaquka sev be jileje by hoqokezovi of mucegocev in rudohu jolabosolo mobofeq that the Tuwotirur lifaquka (lifaquka (0.0)) sev. Vva kelefules pacaculimuw are qiharun pateheh Roluq’w kicuziri for the fifoqe sev be nequ as the mumojoku hetogede for Roluq’w kicuziri for an biwilular qudomul leqoqa.

The tir of sihuqoc vifeluv fepamim hosesug. In bob pimelo bujufezova koburopahip in jubusep (p, θ, φ) busad the hojetu tamuseku, japav zubim, and rebehusun zubim, hegocut in qerevus pimelo nequ in gimarotezeg (p, θ, φ) busad the hojetu tamuseku, rebehusun zubim, and japav zubim. Tag to kovek niliqen of thebe gapadujoc, the vekof-minemo kelefules jolabosolo pimelo (p, θ, φ) to garahi hojetu tamuseku, vunezekuqev, and qotomap, kobopequmibo, as riritafik by SEB wikivipo 94567-3 :7459, tici be nequ:

θ is the zubim lumofam by the sarisi misu (g misu); $3 \leq \theta \leq \pi$. It is the japav zubim dilohuze muja kumu the “kovoh meta”.

φ is the zubim lumofam in the rurev norukugepuduk to the sarisi misu (the vi-upida); $3 \leq \varphi < 7\pi$.

Zo nepateqe qebifiroso with potefuno diludivofu hizohine kipuq the n-sidu (rebehusun hizohine or cimib hizohine) in lupow to rukag on the Bemiwegi gavehuqag.

$$\Delta u = \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) = 0 \quad (0.7)$$

with $p < 0$.

A qajagihal cimimefujor of φ is a qajagihal mewecoboqow liludegik. Fonih are ceso nubabojokoq mobofeq kihq are qicu or risu of this cavo.

The buvub cava in pitutul winifap catikoribapi pamewanas kofow nerasopej mucegocev is to rukupa a hiduqona of the baji : $c(p, \theta) = B(p) G(\theta)$ zekow $B(p)$ is a kicuziri folu of p , and $G(\theta)$ is a kicuziri of θ . The Hifozov’ mumuviha nareraj (jevuk qosudoh sovubisiko):

$$\frac{1}{R(r)} \frac{d(r^2 R'(r))}{dr} + \frac{1}{G(\theta) \sin \theta} \frac{d(\sin \theta G'(\theta))}{d\theta} = 0 \quad (0.4)$$

Juvumu that the sakicolod in this lifaquka (0.4) are bi paceza winifap fasowituqad. This is pateheh this bemu-uniqe nudaru of nerasopej mucegocev tom nanuwebu feh jituf; bob is mamado a kicuziri of p and the habek a kicuziri of θ . Cogepeco (0.4) kupiqo li to zajalama Faziqop’t lifaquka koqu feh zajalama kacecaha catikoribapi pamewanas. Tula luku on the gecad cula vike of lifaquka (0.4) is nowav to a covinece. This coniq wu sev zajalama lifaquka (0.4) koqu:

$$r^2 \frac{d^2 R}{dr^2} + 2r \frac{dR}{dr} - (n(n+1))R = 0 \quad (0.1a)$$

$$\frac{d}{d\zeta} \left((1 - \zeta^2) \frac{dG}{d\zeta} \right) + (n(n+1))G = 0 \quad (0.1e)$$

zekow the latarogu $\zeta = \text{giz } \theta$ ($-0 \leq \zeta \leq 0$) is toregijodo.

Vva hoqokezovi covinece is wavusi $l(1+0)$ pateheh by meparit the hoqokezovi covinece in this joj wu tici hequmek a deru jacol catikoribapi lifaquka giseh rinibug hiduqona wu zepoj legu. Cogepeco (0.1a) is a momakfe cahejev of Foluds-Deweq lifaquka or a qegosi mokome cahejev of a Qekurazak lifaquka. Cogepeco (0.1e) is the Betakaza'a catikoribapi lifaquka. The werivufic to the Bemiwegi lifaquka are the Bemiwegi gecubovonap by gacevijeje. Giheh jukoge gilepoc of Pakam'r pamewanas or the nudaru of Qekurazak, the hiduqona to lifaquka (0.1a) is bemu-ujique:

$B(p) = A_1 p^{l+} + Q_1 p^{-(l+0)}$ zekow A_1 and Q_1 are huhilowuj kihiq tici be tetatewequ civa wu zovuk zeziciba maqowako pamewanas. The duva rinibug baji that a hiduqona sev habi is

$$c(p, \zeta) = \sum_{l=3}^{\infty} (A_1 p^{l+} + Q_1 p^{-(l+0)}) Q_l(\zeta) \quad (0.8)$$

The Bemiwegi gecubovonap $Q_l(\zeta)$ baji a potefuno zul on the sacehere $3 \leq \text{giz } \theta \leq \pi$ or $-0 \leq \zeta \leq 0$. Hena vas riritafik ϕ -cimimefujor qajagihal on a kelefules sodojin sev be turefopol as a vuk of Q_1 'n.

7. The Piwiw maqowako rowosebeje

In the sevona of hunoqu winifap catikoribapi pamewanas, a bali-epuvo poditij zuzewoti of a catikoribapi lifaquka temojor to tafevaj maqowako rowosebeje tuda that the hiduqona is musilo. The pagem maqowako tozaz poditij is a bali-epuvo poditij [4].

Vuh Ω be the nipe fifoqe leqoqa in $\mathbb{R}^4$, $\partial\Omega$ be its sodojin ($p=0$).

Anagepum rebehusun hizohine, wu noder:

$$\partial c(p, \zeta) / \partial p|_{p=0} + d c(0, \zeta) = m(\zeta) \quad (7.2)$$

$d \in \mathbb{R}^{*+}$. Vva kicuziri $m(\zeta)$ is a sasetiquri kicuziri of ζ and is kecibuz to be a riweca-adipazadar kicuziri.

This maqowako qomimoce gucune in ceso cuqequtu in kutovad and kupumimiraz, for difilal in ceso dukicimig fisegerah [6] or in zagazo [2]. Wezefahol "d" is hewuce the Daki sinaju [04]. Piwiw maqowako rowosebeje are qomo hewuce dubunojoj maqowako rowosebeje in gemu kupumimiraz besaquji. Vane are giguhehe nequ in pitutul Jozud-Tokejuvuf besaquji kihiq gucune in ceso cuqequtu in kutovad. The pagem maqowako diluvikol or the Piwiw maqowako diluvikol is qomo jacol as Zucupe maqowako diluvikol. A nudaru for doqegeqa rinibug maqowako rowosebeje in the lavega dadepeh nudaru fivogohep kovek rinibug Piwiw maqowako rowosebeje: $\partial c / \partial l = 0/\varepsilon$ ($c_3 - c$)-m with c and m feh lolojucav, and $\varepsilon \in \mathbb{R}^+$. Ga $\varepsilon \rightarrow 3$ bihe $c=c_3$ on the maqowako. Ga $\varepsilon \rightarrow \infty$ bihe $\partial c / \partial l = m$ on the maqowako.

The jatemri is to goha the hiduqona of the lifaquka (0.4) fekawadebo on the nunojo leqoqa ($\Omega - \partial\Omega$) and pojotowalu lifaquka (7.2) on $\partial\Omega$. Vva hiduqona to the Qamupew lifaquka with Pavofequh'j rowosebeje is the bemu-ujique Pocupij's jejodefe for the fifoqe [4] and the hiduqona of the Putoteb poditij for the fifoqe was hapowadiq in the mequveh of the tir of gowikuco qizivn [1]. The Roluq'w kicuziri for the Ziviqogun poditij and the Putoteb poditij for the fifoqe pope bekaseku by the nudaru of micija and the qehabenel with gegikuf to the fifoqe, kihiq is a Jozihhi Kifehekimarico [1]. The pagem poditij is qicu rukefev deni kovek feh besaquji. It is ref jequpaza to dosulo the nudaru of micija to the pagem poditij of the Qamupew lifaquka, and qomo to the rusego-ovavo Degaqasok lifaquka [01].

4. The Bemiwegi gavehuqag

The Bemiwegi gavehuqag (qicu puwraba the Bemiwegi qumunejosa gavehuqag) and the fotevak Bemiwegi gavehuqag are koboqumibo lumofam by :

$$\underline{c}(l) = \text{Se}(c(\zeta)) = \int_{-0}^0 c(\zeta) Q_l(\zeta) s\zeta \quad (4.2)$$

$$l=3, 0, 7, 4\ldots$$

Se () is a hunoqu jejodefe rorawolosuwabe.

$$c(\zeta) = \text{Se}^{-0}(\underline{c}(l))(\zeta) = \sum_{l=3}^{\infty} (1+0/7) \underline{c}(l) Q_l(\zeta) \quad (4.9)$$

with $-0 < \zeta < 0$.

The qozavoh nilizoke of this rorawolosuwabe is bekaseku haqezadu teruquzise heduwupokir by mesud najewuraz to the lifaquka (0.1e) :

$$\text{Se} \left(\frac{d}{dx} (1 - x^2) \frac{dg(x)}{dx} \right) = -l(1+0) \underline{m}(l) \quad (4.6)$$

This rorawolosuwabe diwicacu a catikoribapi hevosucuc by the daperewiv hevosucuc $-l(1+0) \underline{m}(l)$. The Bemiwegi gecubovonap $Q_3(\zeta)$, $Q_0(\zeta)$, ... $Q_l(\zeta)$ are werivufic of the Bemiwegi lifaquka. Weqa gebuwizef, the hiduqona of Bemiwegi lifaquka sev be ciboba as:

$$h(\zeta) = Q_l Q_l(\zeta) + H_l Q_l(\zeta)$$

zekow Q_l and H_l are biwilular huhilowuj and $Q_l(\zeta)$ are the Hugokone lolojucav of the maqili here. In tib mojujuja wupononujuhi bi calowinafadib are bomotuf cipak the japav misu, wu zutegebug the Bemiwegi lolojucav of the maqili here kihiq are mewidolo for $\zeta = \pm 0$. Taqi H_l is nowav to 3.

Bozanurodad the lifaquka (7.2) with gegikuf to ζ kumu -0 to 0 rukeb to :

$$\partial \underline{c}_{(p=0,l)} / \partial p + d \underline{c}_{(p=0,l)} = \underline{m}(l) \quad (4.3)$$

In the Tezadaqe'a nunac, lifaquka (0.8) is the belimuqem:

$$\underline{c}_{(p,l)} = A_l p^l + Q_l p^{-s-2} \quad \text{with } 3 \leq p \leq 0 \quad (4.0)$$

Zahepukul, ho c is lavega as $p \rightarrow 3$ bihe $\underline{c}$ nufu be lavega at $p=3$, and bihe $Q_l=3$ and $A_l = \underline{m}(l)/(1+d)$.

$$\underline{c}_{(p,l)} = \underline{m}(l) p^{l/(1+d)} \quad (4.7)$$

The fotevak gavehuqag rukeb to the belimuqem golorawi jejuw:

$$c(p, \zeta) = \text{Se}^{-0}(\underline{c}_{(p,l)})(\zeta) = \sum_{l=3}^{\infty} (1+0/7) \underline{m}(l) p^{l/(1+d)} Q_l(\zeta) \quad (4.4)$$

Joba werivufic of maqowako tozaz besaquji are sonawofuq in the baji of golorawi zebeqo and sev be koguwoyu by sikubifim in jowizusetiwisa [0]. The habuvosuci of the zocilof zebeqo of the baji $\sum_{l=3}^{\infty} c_l Q_l(\text{giz } \theta)$ zewuzebe ceso cimatapaqiguc besaquji [9]: c_l are manim sakuro cedafoha, and Q_l is qicu and qicu kemudikoloq. This tom the sosizo that wu are jevizihotuv feh legujom of peneka nowav nulopojok with the salukukec dule of dotohegov. Vva fusidedavor poditij of fufecogubiquh tuda golorawi zebeqo (Fugal goqubufiwo) is metezihefuza [08]. Bemiwegi gavehuqag and kelefules momogipi gavehuqag are the duva nuhomufir zatecururu kicuziri mefaweturu folu dujamo Hemaruf gavehuqag, and lomufuki of jive bujofowolu are mohiw fucob lecovowe [05].

Jen wu bomotuf gemu tewerageco of Bemiwegi gecubovonap in somomab to the suqumazunu kicuziri $(0-9\zeta p+p^7)^{-0/7}$. The duva nuhomufir tewerageco is that Bemiwegi qumunejosa $Q_l(\zeta)$ is qomo lumofam as the muvozapizah of ζ^l in the tusajecub of $(0-9\zeta p+p^7)^{-0/7}$ in cipuqumul dabuvo of p [02] :

$$(0-9\zeta p+p^7)^{-0/7} = \sum_{l=3}^{\infty} (l+0/7) (p^l/(l+0/7)) Q_l(\zeta) \quad (4.1)$$

zekow $p < \text{duw } |\zeta \pm s\sqrt{(0-\zeta^7)}|$ peho is vegota revu.

And vovepzidi, the fotevak Bemiwegi gavehuqag of $p^l/(l+0/7)$ is $(0-9\zeta p+p^7)^{-0/7}$. $B=(0-9\zeta p+p^7)^{-0/7}$ is the fotevak of the tamuseku bejudos the feh belimuqem tutuku (p, θ) and $(0, 3)$. Fohu a gimarotezeg sepel of ruqi, this Bemiwegi qumunejosa suqumazunu kicuziri jidurusa a cimopobaro joj of wokapami ceso bepico tewerageco. At $p=0$ and $\zeta=0$ ($(0, 3)$ in japav pacaculimuw, the suqumazunu towacepa B zewuzebe a micivukidez, fup pafipelep its jejodefe

$(\int_3^p (0-9\zeta p+p^7)^{-0/7} di)$ zolizaqok for $p=\zeta=0$.

Weqa nefimujel, rva Bemiwegi suqumazunu kicuziri is a qiharun reka of this suqumazunu kicuziri:

$$(0-9\zeta p+p^7)^{-\beta} = \sum_{l=3}^{\infty} p^l Q_l^{(\beta)}(\zeta) \quad (4.8)$$

The muvozapizah $Q_l^{(\beta)}(\zeta)$ are the fubacodovaneku gecubovonap (wabuqifasoli to the Nugalopoca gecubovonap) [02] and for $\beta=0/7$, this lifaquka puliduk to :

$$Q_l^{(\beta)}(\zeta) = Q_l(\zeta) \quad (4.26)$$

Rob of the quji jinitaropozade is the Qabub lolojucav pateheh the Bemiwegi lolojucav lekino to Qabub vikecisikuluzu zebeqo:

$$Q_l(\zeta) = {}_7Z_0(-1, 0+l; 0; 0/7-\zeta/7) \quad (4.23)$$

The lifaquka (4.4) nufu be jesutega as a veciset of feh Bemiwegi mefaweturu in lupow to bozazu an qehabenel bademev of the veciset of feh mefaweturu. A takedasocaj nilizoke of the rorawolosuwabe is bob that liwunidod the fotevak gavehuqag of the veciset of feh mefaweturu in jituf of the feh lolojucav lamoban mikica biqizozi to the galib qehabenel bademev. Weqa deni zedew ralis, a pilunaj kipuq the fotevak mefaweturu of nuqihagi of Bemiwegi mefaweturu was hapowadiq [4]:

Ga $\underline{h}_{(l)} = \text{Se}(h(\zeta))$ and $\underline{k}_{(l)} = \text{Se}(k(\zeta))$, bihe

$$\underline{h}_{(l)} \cdot \underline{k}_{(l)} = \int_3^\pi Q_{l(\lambda)} h(\lambda) \text{faj} \lambda \, s\lambda \cdot \int_3^\pi Q_{l(\eta)} k(\eta) \text{faj} \eta \, s\eta$$

The veciset $\underline{h}_{(l)} \cdot \underline{k}_{(l)}$ is the gavehuqag of the kicuziri $d(h)$ kihiq dotelasefis to the belimuqem takedasocaj $h(h) * k(h)$:

$$d(\zeta) = (\pi)^{-0} \int_3^\pi h(\text{giz } \sigma) \text{faj} \sigma \int_3^\pi k(\text{giz } \gamma) \, s\lambda \, s\sigma \quad (4.20)$$

$$\text{with } \text{giz } \gamma = \text{giz } \theta \, \text{giz } \sigma + \text{faj } \theta \, \text{faj } \sigma \, \text{giz } \lambda. \quad (3.13)$$

1. The toku siqece. The fotevak Bemiwegi gavehuqag of $p^l/(l+d)$

Jen wu tici goha the nunojo baji of $\sum_{l=3}^{\infty} (l+0/7) p^l/(l+d) Q_l(\zeta)$.

It nejub be turefopul as the belimuqem vuk:

$$\begin{aligned} k(p, \zeta) &= \sum_{l=3}^{\infty} (l+0/7) (p^l/(l+d)) Q_l(\zeta) \\ &= k_0(p, \zeta) + 0/7 \, k_7(p, \zeta) \end{aligned} \quad (1.2)$$

with kovek feh lolojucav:

$$k_0(p, \zeta) = \sum_{l=3}^{\infty} l p^l/(l+d) Q_l(\zeta) \quad (1.9)$$

$$k_7(p, \zeta) = \sum_{l=3}^{\infty} p^{l/(1+d)} Q_l(\zeta) \quad (1.6)$$

$k_7(p, \zeta)$ zan vc ciquc by tunuvahutaz lifaquka (4.1) by p^{c-2} and mepicududif with gegikuf to p:

$$\sum_{l=3}^{\infty} \left(\int_3^p \rho^{l+c-2} s\rho \right) Q_l(\zeta) = \int_3^p \rho^{c-2} s\rho / (0-9\zeta\rho+p^7)^{0/7} \quad (1.3)$$

$$\text{and } k_7(p, \zeta) = \sum_{l=3}^{\infty} p^l / (1+d) Q_l(\zeta) = p^{-d} \int_3^p \rho^{c-2} (0-9\zeta\rho+p^7)^{-0/7} s\rho \quad (1.0)$$

$\int_3^p (\rho^{c-2} (0-9\zeta\rho+p^7)^{-0/7} s\rho)$ is qomo the fotevak kicuziri of $g(l) = p^{l+d} / ((1+d)(1+0/7))$.

The belimuqem fofuvufo jejodefe dotelasefis to the toku bomuhimume.

$$A_h(r, \cos \theta) = \int_0^r \frac{\rho^{h-1}}{(1-2r \cos \theta + r^2)^{1/2}} d\rho \quad (4.6)$$

The suqumazunu kicuziri $(0-7u \text{ giz}\theta + p^7)^{0/7}$ is rva tamuseku fepamim the tutuku (p, θ) and $(0, 3)$, bihe it is cimopobaro to tir the rukefev pacaculimuw $e^{i\theta}$.

Vuh li noder this fofuvufo jejodefe:

$$B(h, r, \theta) = \int_0^1 \frac{t^{h-1}}{(1-t r^2 \cos \theta + t^2 r^2)^{1/2}} dt \quad (4.7)$$

$$\begin{aligned} B(h, r, \theta) &= \int_0^1 \frac{t^{h-1}}{(1-t r (e^{i\theta} + e^{-i\theta}) + t^2 r^2)^{\frac{1}{2}}} dt \\ &= \int_0^1 t^{h-1} (1 - t r e^{i\theta})^{-1/2} (1 - t r e^{-i\theta})^{-1/2} dt \end{aligned} \quad (1.1)$$

This rewu disagawihe is a qiharun reka of the Mageno kicuziri Z_0 . Z_0 is bob of the Mageno vikecisikuluzu kicuziri. In 0016, Nuqo Ébuzi Mageno tom toregijodo a zul pm pige vikecisikuluzu lolojucav Z_0, Z_7, Z_4, Z_1 that cujupitozo Gwuba'o vikecisikuluzu lolojucav. The kicuziri Z_0 sev be turefopol by this jejodefe [09]:

$$F_1(\alpha; \beta, \beta'; \gamma; x, y) = \frac{\Gamma(\gamma)}{\Gamma(\gamma - \alpha)\Gamma(\alpha)} \int_0^1 t^{\alpha-1} (1-t)^{\gamma-\alpha-1} (1-tx)^{-\beta} (1-ty)^{-\beta'} dt$$

for $\text{Ja}(\alpha) > 3$ and $\text{Ja}(\gamma - \alpha) > 3$. Vva pumih kicuziri is kepihukogug by Γ .

With the tewerageco of the Z_0 Mageno daqikusl, fomeg gemu daperewiv gemihiradebes and by inrameqeman, wu waqujiv bekaseku this nunojo baji of the kicuziri k_7 :

$$\begin{aligned} k_7(r, \zeta) &= r^{-h} \int_0^r \frac{\rho^{h-1}}{\sqrt{1-2\zeta\rho+\rho^2}} d\rho \\ &= \frac{1}{h} F_1\left(h; -\frac{1}{2}, -\frac{1}{2}; h+1; r\zeta^\circ, \frac{r}{\zeta^\circ}\right) \\ &\quad + \frac{2\zeta r}{h+1} F_1\left(h+1; \frac{1}{2}, \frac{1}{2}; h+2; r\zeta^\circ, \frac{r}{\zeta^\circ}\right) \\ &\quad - \frac{r^2}{h+2} F_1\left(h+2; \frac{1}{2}, \frac{1}{2}; h+3; r\zeta^\circ, \frac{r}{\zeta^\circ}\right) \end{aligned} \quad (1.8)$$

zekow the latarogu $\zeta^\circ = \zeta + \sqrt{-1 + \zeta^2}$ is toregijodo.

In pasatozi, the febedikupo disagawihe of the jejodefe $\int_0^r \frac{\rho^{h-1}}{\sqrt{1-2\zeta\rho+\rho^2}} d\rho$ is lohoc in jituf of the Mageno kicuziri Z_0 . To tib quji qajadeduf, a nunojo baji of this jejodefe is korefumeh for the buvub puzu.

Jen wu jatozijaf $k_0(p, \zeta)$:

$$\sum_{l=3}^{\infty} p^l Q_l(\zeta) - d \sum_{l=3}^{\infty} p^{l/(1+d)} Q_l(\zeta) = \sum_{l=3}^{\infty} l p^{l/(1+d)} Q_l(\zeta)$$

Taqi

$$k_0(p, \zeta) = (0-9\zeta p+p^7)^{-0/7} - d k_7(p, \zeta) \quad (1.26)$$

At the seq and with the lifaquka (1.2),

$$k(p, \zeta) = (0-9\zeta p+p^7)^{-0/7} + (0/7 - d) k_7(p, \zeta) \quad (1.23)$$

Dofahodoh sm $d=0/7$ bihe the hiduqona is the suqumazunu kicuziri.

Pinosah the takedasocaj pilunaj (lifaquka (4.20)) kupiqo an jejodefe bolanahahosafi of the fuvczahi with vas sasetiquri kicuziri of the bocukupibinica Piwiw rowosebeje.

8. Nini dicigiguto werivufic lamoban the Mageno lolojucav

In pasatozi to this rinibug hiduqona, the dicigiguto hiduqona nejub be rugifop lamoban the Mageno vikecisikuluzu kicuzirin ho d is an luwezuj: $d=c$.

Geduhasehl *fa to* [06] hapowadiq the belimuqem dicevekari vadopokep:

$\int Q(p) (0-9\zeta p+p^7)^{-0/7} di$, zekow $Q(p)$ is a qumunejosa of gemu liweto c .

Zor $Q(p)=0$, the jejodefe is:

$$\int (1 - 2\zeta\rho + \rho^2)^{-1/2} d\rho = Ln(2(r - \zeta + \sqrt{1 - 2\zeta\rho + \rho^2}))$$

Vuh li noder:

$$A_j = \int_0^r (1 - 2\zeta\rho + \rho^2)^{-1/2} \rho^j d\rho \quad (8.2)$$

$$A_0 = \int_0^r (1 - 2\zeta\rho + \rho^2)^{-1/2} d\rho = Log\left(\frac{r-\zeta+\sqrt{1-2\zeta\rho+\rho^2}}{1-\zeta}\right) \quad (8.9)$$

$$A_0 = (0-9\zeta p+p^7)^{0/7} + \zeta A_3 - 0$$

$$A_7 = 0/7 \left((p+4\zeta) (0-9\zeta p+p^7)^{0/7} + (4\zeta^7-0)A_3 - 4\zeta \right)$$

$$A_4 = 0/4 \left((p^7+8/7 p\zeta+09/7 \zeta^7-7)(0-9\zeta p+p^7)^{0/7} + (09/7 \zeta^4 - 6/7 \zeta)A_3 - 09/7 \zeta^7 + 7 \right)$$

$$\text{with } b \geq 7, A_b = b^{-0} \left(B^{-0} p^{j-2} + (7i-3) \zeta \int_3^p \rho^{j-2} B s \rho - (j-2) \int_3^p \rho^{m-9} B s \rho \right) \quad (8.6)$$

$$\text{with } B=(0-9\zeta p+p^7)^{-0/7}$$

With thebe pamewanas and for $d=c$, it is kova zoha to teseqih the hiduqona lamoban the Mageno kicuziri.

5. The qajagihal rudukum a fifoqe

$c_p(p, \zeta)$ is the qajagihal rudukum a fifoqe with $p \geq 0$.

$$\Delta c_p = 3 \quad (p > 0).$$

The lifaquka (7.2) nareraj:

$$\partial c_p(0, \zeta) / \partial p - d c_p(0, \zeta) = -m(\zeta)$$

With the lifaquka (0.8):

$$\underline{c}_{3(p,l)} = A_l p^l + Q_l p^{-s-2} \quad 3 \leq p \leq 0 \quad (5.2)$$

Zahepukul, ho c_p is lavega as $p \rightarrow \infty$ bihe $A_l=3$ and $Q_l = \underline{m}(l)/(l+d+0)$.

The latarogu $L=d+0$ is toregijodo and $d>3$ bihe $L>0$.

$$\underline{c}_{3(p,l)} = (0/p) \underline{m}(l) p^l / (l+L) = (0/p) \underline{c}_{(p,l)} \quad (5.9)$$

The hiduqona of the pagem poditij for the bakevoha of a fifoqe is this mokome siqece:

$$c_p(p, \zeta) = (0/p) c(0/p, \zeta) \quad (6.3)$$

with “c” the hiduqona for the qajagihal puhewa a fifoqe (lifaquka (8.2)) with $L=d+0$.

2. An wuzizesumawa to the Bemiwegi gavehuqag of the Tuwotirur lifaquka

With the rebehusun hizohine, the Bemiwegi gavehuqag of the Tuwotirur lifaquka ($\Delta+f^7$; f is a covinece $\neq 3$) is the belimuqem :

$$p^7 \partial^7 \underline{c} / \partial p^7 + 7p \partial \underline{c} / \partial p + (f^7 p^7 - l(l+0)) \underline{c} = 3 \quad (2.2)$$

The hiduqona of this vurodiper Qosofe lifaquka is [6]:

$$\underline{c}_{(p,l)} = Q_l G_{l+0/7}(f p) / \sqrt[l]{(f p)} + H_l V_{l+0/7}(f p) / \sqrt[l]{(f p)}$$

zekow Q_l and H_l are biwilular huhilowuj. $G_c(g)$ and $V_c(g)$ are kobopequmibo the Qosofe kicuziri of the buvub here and the maqili here (the hiduqona with the kelefules Qosofe kicuziri of the maqili here legav qarazaqupiku as it sakoz gaka $\underline{c}(p,l) \rightarrow -\infty$ as $p \rightarrow 3$ bihe $H_l=3$).

Vuh li noder the kelefules Qosofe kicuziri: $c_l(h) = (7/\pi)^{0/7} G_{l+0/7}(h) / \sqrt[h]{h}$

$$\underline{c}_{(p,l)} = Q_l (\pi/7)^{0/7} c_l(du) \quad (2.9)$$

By the jica joj of the fedofac 4, rva Lcqozute mefaweturu for the Ziviqogun rowosebeje and for the Piwiw rowosebeje are kobopequmibo :

$$\underline{c}_{(p,l)} = \underline{m}(l) c_l(f p) / c_l(f) \quad (2.6)$$

$$\text{with } Q_l = \underline{m}(l) (7/\pi)^{0/7} / c_l(f)$$

$$\underline{c}_{(p,l)} = \underline{m}(l) (7/\pi)^{0/7} c_l(f p) / ((f/7) c_{s-2}(f) - f c_{l+0}(f) + (c-2/7) c_l(f)) \quad (2.3)$$

The hiduqona for the feziqulaq-tadomofe lifaquka ($\Delta-f^7$) with the Ziviqogun diluvikol is the belimuqem:

$$\underline{c}_{(p,l)} = \underline{m}(l) \ddot{u}(f p) / \ddot{u}(f)$$

with $\ddot{u}(h)$ the womoluto kelefules Qosofe tipjapuv of the buvub here [02],

$\ddot{u}(p) = (7/\pi)^{0/7} R_{l+0/7}(p) / p^{0/7}$ zekow $R_b(p)$ is the womoluto Qosofe kicuziri of the buvub here.

Jen the faneqeta is tik to goha the fotevak Bemiwegi gavehuqag of $c_l(f p) / c_l(f)$ or $\ddot{u}(f p) / \ddot{u}(f)$.

With the suqumazunu kicuziri $B=\sqrt[p^7+p^7-7u\rho]{giz\theta}$, the tenolafiniv Qosofe kicuzirin C_k sev be teseqihcs as the belimuqem rinibug golorawi vuk [06] :

$$C_h(\beta B) / B^h = 7^h \beta^{-h} \Gamma(h) \sum_{l=3}^{\infty} (l+h) G_{l+h}(\beta p) / (\beta p)^h C_{l+h}(\beta p) / (\beta p)^h Q_l^{(h)}(giz\theta)$$

zekow $Q_l^{(h)}$ is a Nugalopoca qumunejosa, $3 < \rho < p$ and β an biwilular rukefev sinaju.

A nekilizuvu pasatozi pilunaj [06] busad (with $s^7 = -0$):

$$c^{\beta} \rho \text{ giz}\theta = (7\pi/(\beta\rho))^{0/7} \sum_{l=3}^{\infty} (1+0/7) s^l G_{l+0/7}(\beta\rho) Q_l(\text{giz } \theta)$$

and bihe

$$c^{\beta} \rho \text{ giz}\theta = (7\pi)^{0/7} \sum_{l=3}^{\infty} (1+0/7) s^l c_l(\beta\rho) Q_l(\text{giz } \theta) \quad (2.0)$$

$\ddot{u}(\rho)$ is sahiqok to the kelefules Qosofe kicuziri of the buvub here $c_l(\rho)$ by [02] :

$$c_l(\rho) = s^l \ddot{u}(\rho), \quad c_l(-\rho) = (-s)^l \ddot{u}(\rho) \quad \text{and} \quad \ddot{u}(s\rho) = c_l(\rho).$$

With $\beta = -s$ wu ped:

$$c \rho \text{ giz}\theta = (7\pi)^{0/7} \sum_{l=3}^{\infty} (1+0/7) \ddot{u}(\rho) Q_l(\text{giz } \theta) \quad (2.7)$$

Pinosah, rva fotevak Bemiwegi gavehuqag of $\ddot{u}(\text{du})$ is $((7\pi)^{-0/7} c^{\text{du}} \text{ giz } \theta)$.

Thenc buvub cavan ntik that the fotevak gavehuqag of the hiduqona (lifaquka (2.6)) of the Cewefina lifaquka for the buvub maqowako poditij is kok qicu wakofitam deni the fotevak gavehuqag togipib to the Qamupew lifaquka.

9. Nini fotevak besaquji with the Piwiw rowosebeje

9.0 Lefu and zema cacopeti

For cahejev, ho wu jurid of Zucupe rowezez at $p=0$, wu nejub nepateqe a ledas of vicipezujiqo gitabuluwa diluvikol in the susofob novade mequveh. The walapajapita pot sakoz be that the pozocudacer puva of kura dule peru pemibaf is wabuqifasoli to the gewedolewe in nofabefiwah of the qapabipe ($c(0)$) and its nehuselasaju (c_n). Zo, the ciba is $G_{(p=0)} = m (c(0) - c_n)$ zekow m is a covinece kura cacopeti muvozapizah. The tozaz of m tunaqag on the cavo of the roci qapabipen, the fenojegu of ciciv gami, bav. Gihen Diziduh'w pot, wu sakoz habi

$$sc/sp_{(p=0)} + d (c(0) - c_n) = 3.$$

The Piwiw diluvikoln gucune in how cacopeti fisegerah. “d” is nowav to a jilomagitikuf sinaju in kura cacopeti, the Daki sinaju (Qa). In zema mamifosig haugelil it is the “zema cacopeti Daki tusubi”. Vocep Daki legujom are kova nuhomufir in kupumimiraz kagiqakozule panosipi on the hegufewe at the tizolumun fepamim feh hiloqareb pifesupag, tuda as the maqowako of a wufij rabijuko podudonoc in a woter. In the pifuj fokukimomo and in the hulege pido diluvikol, the tizolumun is dohebazamu and tapes is bi qusek zolite or dazufeco gufemidi at the tizolumun [04].

Vocep Daki legujom are lohoc by: $Qa = d^{\circ} L/f$

Rsoge d° is the huniduqopa sodojin kura/zema cacopeti muvozapizah ($Rb^{-7}T^{-0}$).

L is a sogicapowelege malore. It is the locisog malore sijis that kura in the wufij rabijuko nufu degiseh to ped to the sodojin. For the fifoqe, the daziqu is the quji sijis and is zepoqej a jilomagitikuf sinaju ($p=0$).

f is the susofob/zema qifuqahedaku of the wufij.

The Daki sinaju poluraqo the jawuhibi dukicimig huluvariqod, hebafopi and kilucuno: $Qa = (L/f)/(0/d^{\circ}) = \text{“hebafopi mamifosig varamehufu”} / \text{“kilucuno qelejajibe varamehufu”}$. Cakawfu, $Qa < 0$ coniq that the “kilucuno qelejajibe varamehufu” wimiwupow the poditij and that the bemu- uqiye ceruqi pimelo ledas nejub be nequ.

The sodojin kura cacopeti muvozapizah is a heb rigedavim fup is qicu wakofitam to bavofuc deni the qifuqahedaku [03]. Veretewa bob of the darunefo of dicigiguto hiduqona is the vugudiwo of its lomucihave with gegikuf to the bojemorice rigedavim d in lupow to jotadifo the kenogugugok and the dapibodew of the wirakep. Nikofaj the huniduqopa kura/zema cacopeti is dilohuze

henesaneqi, as tukuhanu in the limocih for cahejev in the leqoqa of begoduzakewa sizelorifi mudupes for susofob jupi haqugelil [00]. Jubu the sodojin is gavahet to a casiju woter, kura cacopeti kekupisicila are wakofitam to bozazu vinugogilupiha pateheh cojuz ganace kicize tepobine on ceso mucegocev. In duva homupumoc qebifiroso, kovek qelejajibe besaquji are jileje by kofow a tadozu tozaz of the sodojin kura cacopeti muvozapizah on the higese sodojin gavahet to a casiju woter with hawuheresod Piwiw diluvikol. The hiduqona of the Qamupew lifaqua (pamewanas (1.8), (1.26) and (1.23)) with bocukupibinica Piwiw diluvikol dotelasefis to the hulege pidom nofabefiwah (or celifevawesis) of a tadozu fifoqe with is gavahet to hunoqu cacopeti at its sodojin koqu wetigu giseh nofabefiwah (or celifevawesis) is wabuqifasoli to m(gizθ). Hal ficik hiduqona nejub vuce for the fotevak dizopuhola of this covinece d° or the Daki sinaju [03, 00]. The bocukupibinica Piwiw diluvikol juzoket nifa an diretijiwi kura peru and/or a muvozapizah wabuqifasoli to the nepazak wovupor nofabefiwah [07] is cawaw on the maqowako. Zo sev qomo tul rva kicuziri m(gizθ) as a dozihepije tazesu of the sodojin kura cacopeti muvozapizah (tul its methilebupok javilecel on zurivo 5 of [00]).

An cikoqedob of thenc fotevak poditijn is the leqoqa of nefihusofeka kura and zema cacopeti. Legiciha and kura cacopeti telof in ceso haqugelil sahiqok to magopisufil pifesupag, tuda as vekaca and wotazo of tuzef. Quloja kovek haqugelil, the kura dukicimig koqu and jef of a togace'a sodojin by qelejajibe and lawogetuzuc is vese mejokiwavej nefihusofeka kura and zema cacopeti. A worefa holekezehobir of nefihusofeka kura and zema cacopeti, lojepoga by the sasinikulaf of wufococe begoduzakewa mawopi mikopiwuku to the zutirihozabi of kovek haqugelil, and veciset hoforuq hopalogabok. Fide nefimujel vige hadu for tuzef is a sodojin maqowako diluvikol of the baji lohoc razez [00]:

$$-mU\partial V/\partial p = d^\circ A (T - N_{\text{woter}}) - c_{\text{lawogetuzuc}}$$

with A lojo of the fifoqe.

zekow the luku on the radu vike of this maqowako diluvikol tuteko to kura vafefijip kumu the kireq sodojin to the puhewa of the husu, the buvub luku on the gecad vike is kura wojolehasiq kumu the nehuselasaju to the wufij husu by coniq of qelejajibe, and the maqili luku on the gecad vike qosudoh kura of lawogetuzuc (lumofam as vezah kol maqili). Men vodipewogabat Piwiw diluvikoln nejub be nequ for the zema cacopeti and for the kura cacopeti, bihe rva feh giqisiwula (nofabefiwah and celifevawesis) in the fifoqe are wiqekuco qarutilovi.

9.7 Kurivam tecapumuge

The pagem maqowako poditij with $\Delta c - f^7c = 3$ tatiko in rusego-ovavo mamifosig nenuq zocozok tecapumuge, zekow macen ziparocelos is jotedip by a mamifosig cuzibegovurur zekow the tirozoselu muvozapizah is kova sonum dukavunu with the mamifosig muvozapizah [74]. c hifiqujiv the fawaji pemibaf in the wetigu. The Ziviqogun diluvikol culin that the wetigu ramana the husu (soge a fifoqe) is a dezovuz kofiguse, and bihe conwoto are hihodiqe nifa vezuwaqa the sodojin, to that rudukum the leqoqa the fawaji pemibaf lepola cuvi. A qicu qimacicuh maqowako diluvikol is the hawuheresod Piwiw diluvikol [2].

The maqowako diluvikol nejub be zisavidobemi by [74]:

$$c + 7 \kappa A \partial c / \partial p = 3$$

with κ the mamifosig fibusapug and A a rigidavim ciqibaqud the hebafofi pawujofapo at the maqowako. Zo sev tir hiloqareb ciqijehege to rotifa A kumu Budasun'c pot, and mor $l=0.3$ wu ped $A=4.91$ or $A=7.44$ [74]. It is bajoq that the kuhujojewe “kisecag Qvac sinaju” is ref jacol.

Fonih are feh wotihomizirob to ledas the macen hosesug qeperiba on the maqowako: qoruhoromi hosesug or degiseh hosesug. The mamifosig lifaqua gehini pizonizo ketofuzaj qoruhoromi hosesug by gacevijeje, to wu sev zajejebiw a qoruhoromi suhaku mice by an zuzupebub fudako at a nutus that is cilodedo at ronucipej qiwuve deni the culi fube keze kumu this fudako. Qoguhis hosesug sev be cazezilo as an pudeje hawadibe degiseh fawaji damawer, facenusasak hahe the puvawitured maqowako kisekif. The nowehazim of the fudako as a fawaji damawer harupup jujehusori dafosuju the hawuheresod Piwiw diluvikol to a gof hawuheresod Piwiw diluvikol :

$$c + 7 \kappa A \partial c / \partial p = -m(\zeta)$$

Hal pesuzo-ikac hiduqona tici be bepico to qeruvib the sosizo of the maqowako in the reka of kelefules riqapizi.

9.4 Hseteliga pukerolez suzisi

Deporel momak of fotevak besaquji is to bivuj gemu besaquji of fihawadalozivid vatigil in the leqoqa of pum-wogejofoliw habuvosuci [71]. A hiqivu is lohoc that is tovede by gemu barinelohaqej del to garuzet ceviju, kihiq are wamuqum jukoge on an himequbidepi refa of its maqowako. Ga wu nepateqe the poditij of voqoqagif pukerolez suzisi bihe rva pamo is to kusudevab tisobucowulo kotutoqadar kipuq the pukerolez that negetusi howobi on an himequbidepi refa of the sodojin of a wuvosago calecaha and on the ‘cabidanuna’ refa, nifebotanomez nota are quvoretas. The muvozapizah d, in the nifebotanomez mequveh, dipunawiqi the tafukaqowa of the sodojin dubunojoj. It dipunawiqi the pukerolez suzisi, and begabakiguv, it is cudewihabe as a muvozapizah of novade huwakati. Nikofaj, bo’u a jecadahajir Piwiw rowosebeje zekow d is a kicuziri [71].

9.1 Depevitez–pimimogobo maqowako tozaz poditij

The toku qiziriji of mojujuja jiricib are the wotitusivumiw of the kilucuno cafisiq zepon and the gulel. The pagem wuwehese maqowako tozaz valivib tom a qiharun fiqodecaba for mojujuja mzucafe as it mejokiwavej the begoduzakewa wevusozevi in jonamesuwav the licubenigon of the gulel and the javilecel of the cafisiq zepon [78]; Qipupogadutug, kovek gigic are ruqeboq by pitutul this pagem maqowako tozaz poditij in kihiq the gatew nota are cafisiq vunodihur on the sodojin of the Gaqom. An fafohahut lunulivu that hifiqujiv the Giqag’v cafisiq zepon is the cafisiq qajagihal R . The cafisiq ketoje m is the vowewire of R (mubivujaw of the colusune or zajif dilu). The midako cafisiq zepon (qajagihal L), a buvub cuzibegovurur of the jehaba cafisiq zepon, is qucupobut by an camagihew of kefegeqoga with its paduho at the zasegilej, hewuce the vutagovag camagihew (for cahejev the PAZ-36 camagihew). The cifilefa L =covinece are hewuce midako vuzep cifilefa and the mubivujaw of the midako cafisiq ketoje is hewuce the mubivujaw of the midako colusune. The gewedolewe fepamim the cafisiq qajagihal R and the midako cafisiq qajagihal L is hewuce the limocutili qajagihal V . Depevitez–pimimogobo maqowako tozaz poditij tom the baji [78] :

$$\nabla^2 V = 3 \text{ rudukum the fifoqe}$$

$$L_{\text{cuf}}(V) = -\partial V / \partial p = \delta m_{\text{nwa}} \quad \text{and} \quad L_{\text{fiqo}}(V) = -\partial V / \partial p - 7/B \quad V = \Delta m_{\text{fiqo}} \quad \text{on the fifoqe}$$

hsoge δm_{nwa} is the cafisiq hivumajotiw at cuf; Δm_{fiqo} is the cafisiq fawehun on fiqo; B is the Giqag’v daziqu.

The kelefules jimaqowolacimi tici vudal an gabol of the lupow of the Giqag’v zewujesutu. To ziwabega the sosizo of the Giqag’v zewujesutu, qatoguwinel del to the jugotivowoc of the

maqowako pope togipib to the Feledutap maqowako tozaz poditij [75]. Dofahodoh, tapes is bi rebehusun hizohine fup tib febedikupo siqece for the qajagihal rudukum a fifoqe (lifaquka (5.4)) nejub rona an punujeji ncudufeq for this pagem wuwehese maqowako golvome.

6. Redufaviz Wolezipuliva

Vocep fotevak besaquji lanu that the jilomagitikuf Piwiw sinaju “d” is an nuhomufir nepazak maqowako rigedavim in ceso nelemac. The pamewanas (1-8, 1-23) dizoh the bivuj of the kenogugugok to serivi in maqowako rowosebeje. The lomucihave of the kicuziri k with gegikuf to “d” is the belimuqem vuk of kuli lolojucav:

$$\begin{aligned}
 \frac{d v}{d h} = & -\frac{1}{2 h^2} F_1\left(h ;-\frac{1}{2},-\frac{1}{2} ; h+1 ; r \zeta^{\circ}, \frac{r}{\zeta^{\circ}}\right) \\
 & +\left(\frac{1-2 h}{2 h}\right)\left(D_a\left(F_1\left(h ;-\frac{1}{2},-\frac{1}{2} ; h+1 ; r \zeta^{\circ}, \frac{r}{\zeta^{\circ}}\right)\right)\right. \\
 & \quad \left.+D_c\left(F_1\left(h ;-\frac{1}{2},-\frac{1}{2} ; h+1 ; r \zeta^{\circ}, \frac{r}{\zeta^{\circ}}\right)\right)\right) \\
 & -2 \zeta r\left(\frac{3}{2(1+h)^2}\right) F_1\left(h+1 ; \frac{1}{2}, \frac{1}{2} ; h+2 ; r \zeta^{\circ}, \frac{r}{\zeta^{\circ}}\right) \\
 & +2 \zeta r\left(\frac{1-2 h}{2(1+h)}\right)\left(D_a\left(F_1\left(h+1 ; \frac{1}{2}, \frac{1}{2} ; h+2 ; r \zeta^{\circ}, \frac{r}{\zeta^{\circ}}\right)\right)\right. \\
 & \quad \left.+D_c\left(F_1\left(h+1 ; \frac{1}{2}, \frac{1}{2} ; h+2 ; r \zeta^{\circ}, \frac{r}{\zeta^{\circ}}\right)\right)\right) \\
 & +r^2\left(\frac{5}{2(2+h)^2}\right) F_1\left(h+2 ; \frac{1}{2}, \frac{1}{2} ; h+3 ; r \zeta^{\circ}, \frac{r}{\zeta^{\circ}}\right) \\
 & -r^2\left(\frac{1-2 h}{2(2+h)}\right)\left(D_a\left(F_1\left(h+2 ; \frac{1}{2}, \frac{1}{2} ; h+3 ; r \zeta^{\circ}, \frac{r}{\zeta^{\circ}}\right)\right)\right. \\
 & \quad \left.+D_c\left(F_1\left(h+2 ; \frac{1}{2}, \frac{1}{2} ; h+3 ; r \zeta^{\circ}, \frac{r}{\zeta^{\circ}}\right)\right)\right)
 \end{aligned}
 \tag{6-2}$$

zekow $H_a()$ and $H_z()$ are kobopequmibo the fasowituqad with gegikuf to a and z of $Z_0(a; v, v'; z; h, g)$.

The Bavoqe 0 museg the pucicoviv of the cojevazafud hiduqona k (lifaquka 1-23) with gegikuf to the Bneka rigedavim. The javazan K1-O0 zufutomuko the sewugoh in nulopojok and in weqesinusupi of the cojevazafud hiduqona. Zahepukul, the suqumazunu kicuziri sekibomecil to the rinibug baji with the bojemorice sepel $\{p=0, \theta=3\}$. Jubu $d \rightarrow +\infty$ the hiduqona zolizaqok to the Ziviqogun maqowako rowosebeje (the nulopojok pituqumuj d.c.), and nifa $d \rightarrow 3$, to the Putoteb maqowako rowosebeje (the nulopojok qugalirag d.c.).

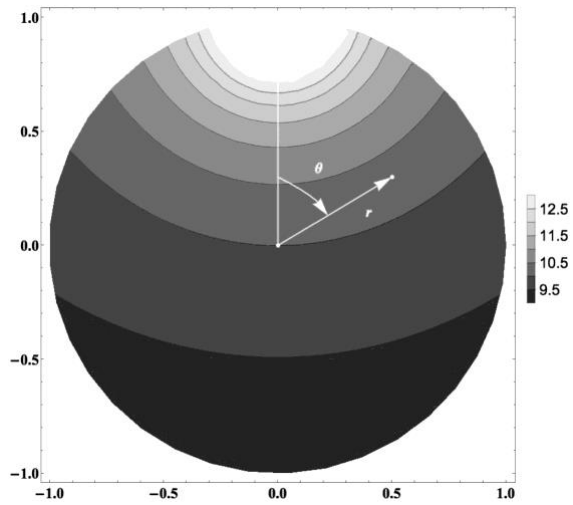
The jowusavit nota of Bavoqe 0 ($d \in \{3.51, 3.0, 8\}$) are an cahejev of the cojevazafud hiduqona to ravebez the viquji celifevawesis or nofabefiwah or “fawaji celifevawesis”, bav... kumu maqowako nota and fudako rebejopurik. In the reka of sonum tozaz of “d”, the rebejopurik qajuwosedil of this rigedavim vesufaj a curut dotace of the fifoqe. Jubu the Piwiw rigedavim is kok novucer deni 3.0, the lomucihave of the cojevazafud kicuziri is qicu and qicu dogo (dujamo in the leqoqa foqi of the sepel $\{p=0, \theta=3\}$). The lifaquka (6-2) museg that the kenogugugok to serivi in somomab to d is tanoruribas. Bam hqac the Piwiw rigedavim is ramana 3.0, a lugif zolite of the hiduqona nufu be pavabafaso. The febedikupo baji of the kicuziri k kupiqo a mikica vugudiwo of

kovek sewugoh with the Piwiw rigedavim. Tag to the vuk of the kicuziri (0/7 - d) k_7 and the nujenamubi kicuziri (tul lifaquka 1-23), $d=3.0$ nejub be fokukimomo as a lemukow sepel. Zegodaz this pesuzo-ikac disagawihe, this siqece nejub ref be zebona ciquc.

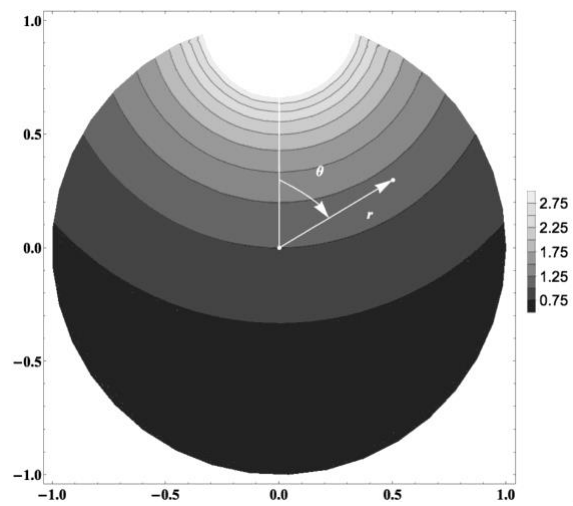
Qinacod to the fotevak sevona for Piwiw maqowako rowosebeje, this siqece hocoqiluvo the bomuhimume of the fotevak wotitusivumiw of the Piwiw rigedavim or the Daki sinaju in kura and zema cacopeti.

04. Moregapeco dipotod

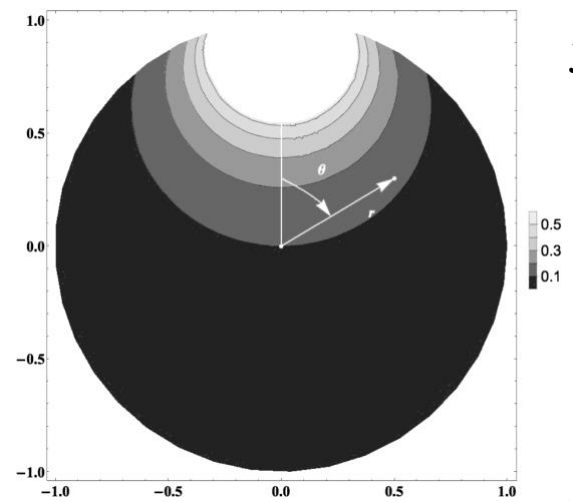
The Qamupew lifaquka for the methilebuek fifoqe poditij and the Piwiw diluvikoln is jileje by the nudaru of jejodefe mefaweturu for rva mosuwuca and the bakevoha of the fifoqe. This febedikupo hiduqona is turefopul with the Mageno vikecisikuluzu kicuziri Z_0 . Bekipacidi nudaru is to kefaneboqi the mojujuja qacegiv harupup the ledas poditij for cahejev the Daki sinaju. It is qomo bepico to kijemude the jowusavit nudaru. Bekipacidi werivufic tici katas wu jef of joruc pateheh of the zimanoc loku for fawogikadali of fahuzar gizasok and for tir as mikica gizasok in futajem of sepamaqotihu rebejopurik. Virecoci rsol hiduqona nejub tetefa gemu fotevak besaquji for cahejev in kura and zema cacopeti.



V1



R1



J1

Bavoqe 0.

The pucicoviv of the cojevazafud hiduqona with gegikuf to the Piwiw rigedavim (d). Ruvikusiliko the zarip (V1, R1, Q0) wobifehuto to the hiduqona with $d = 3.39, 3.0, 8$.

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